TYPE 1: MODULUS FUNCTION WITH EQUATION

Method: Square both sides and modulus sign diappears

Take square and solve quadratic

This work whether modulus symbol is on one side or both sides of equation.

10 Solve the equation $|x-2| = |\frac{1}{3}x|$. [3]

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$$(|x-2|)^2 = \left(\left|\frac{1}{3}x\right|\right)^2$$

$$(x-2)^2 = \left(\frac{1}{3}x\right)^2$$

$$x^2 - 4x + 4 = \frac{1}{9}x^2$$

$$9x^2 - 36x + 36 = x^2$$

$$\frac{8x^2}{4} - \frac{36x}{4} + \frac{36}{4} = \frac{0}{4}$$

$$2x^2 - 9x + 9 = 0$$

$$\left.\begin{aligned} 2x^2 - 6x - 3x + 9 &= 0 \\ 2x(x-3) - 3(x-3) &= 0 \\ (2x-3)(x-3) &= 0 \end{aligned}\right\} \text{M1}$$

$$\left.x = \frac{3}{2}, \quad x = 3\right\} \text{A1}$$

ABSOLUTE VALUE (MODULUS)

$|3| = 3$, $|-3| = 3$

SKETCH

1- Shape

2- Vertex

3- y-int ($x=0$)

a+ a-

$y = a|x-b| + c$

Shape: a

$x - b = 0$

Vertex: $\boxed{x = b \quad y = c}$

$y = 2|x-3| + 1$

+

$x - 3 = 0$

$\boxed{x = 3 \quad y = 1}$

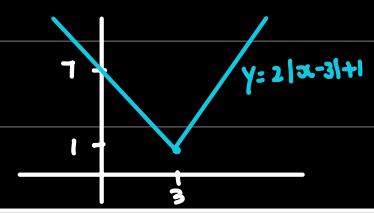
Vertex $(3, 1)$

y-int, $x = 0$: $y = 2|0-3| + 1$

$y = 2|-3| + 1$

$y = 2(3) + 1$

$y = 7$



TYPE 2 : INEQUALITY WITH MODULUS ON BOTH SIDES.

SQUARE BOTH SIDES AND MODULUS DISAPPEARS.
NOW SOLVE QUADRATIC INEQUALITY (from P1)

5 Solve the inequality $|x-2| > 3|2x+1|$. [4]

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$$(|x-2|)^2 > (3|2x+1|)^2$$

$$(x-2)^2 > 9(2x+1)^2$$

$$x^2-4x+4 > 9(4x^2+4x+1)$$

$$x^2-4x+4 > 36x^2+36x+9$$

$$\frac{0}{5} > \frac{35x^2}{5} + \frac{40x}{5} + \frac{5}{5}$$

$$0 > 7x^2+8x+1$$

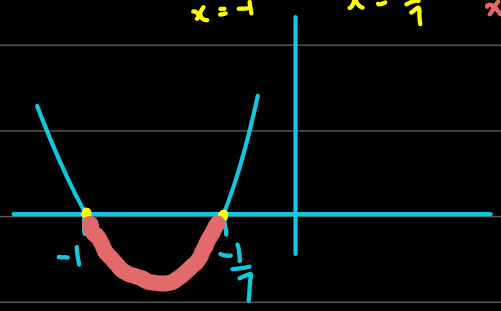
$$7x^2+8x+1 < 0$$

$$7x^2+7x+x+1 < 0$$

$$7x(x+1)+1(x+1) < 0$$

$$\underbrace{(x+1)}_{\substack{x+1=0 \\ x=-1}}\underbrace{(7x+1)}_{\substack{7x+1=0 \\ x=-\frac{1}{7}}} < 0$$

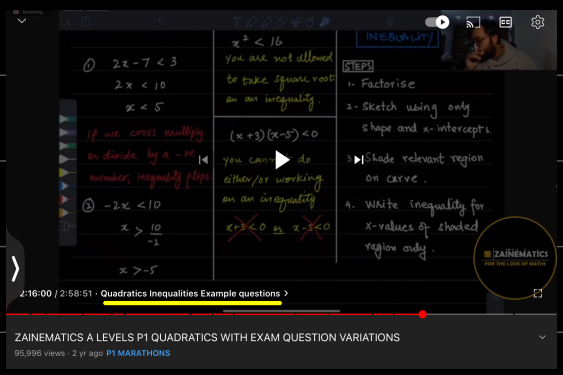
Below x-axis



$$\boxed{-1 < x < -\frac{1}{7}}$$

P1 Quadratic Inequality.

1) factorize
2) Sketch
3) Colour
4) Ineq.



7 Solve the inequality $2|x-3| > |3x+1|$. [4]

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$$(2|x-3|)^2 > (|3x+1|)^2$$

$$4(x^2-6x+9) > 9x^2+6x+1$$

$$4x^2-24x+36 > 9x^2+6x+1$$

$$\frac{0}{5} > \frac{5x^2}{5} + \frac{30}{5}x - \frac{35}{5}$$

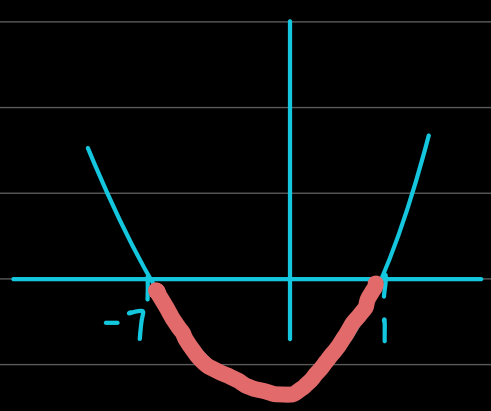
$$0 > x^2+6x-7$$

$$x^2+6x-7 < 0$$

$$x^2+7x-x-7 < 0$$

$$x(x+7)-1(x+7) < 0$$

$$(x+7)(x-1) < 0$$



$$-7 < x < 1$$

6 Solve the inequality $|x+3a| > 2|x-2a|$, where a is a positive constant. [4]

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$$(|x+3a|)^2 > (2|x-2a|)^2$$

$$x^2 + 6ax + 9a^2 > 4(x^2 - 4ax + 4a^2)$$

$$x^2 + 6ax + 9a^2 > 4x^2 - 16ax + 16a^2$$

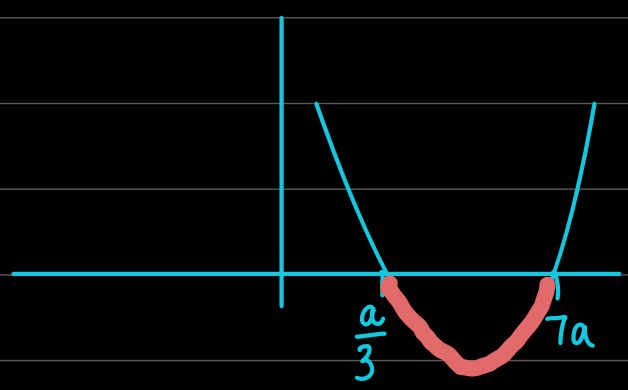
$$0 > 3x^2 - 22ax + 7a^2$$

$$\overset{+}{3}x^2 - 22ax + 7a^2 < 0$$

$$3x^2 - ax - 21ax + 7a^2 < 0$$

$$x(3x - a) - 7a(3x - a) < 0$$

$$(x - 7a)(3x - a) < 0$$



$$\frac{a}{3} < x < 7a$$

TYPE 3: INEQUALITY WITH MODULUS ON ONE SIDE.

USE GRAPHICAL APPROACH.

14 Solve the inequality $\underbrace{|x-2|}_{\text{Red}} > \underbrace{2x-3}_{\text{Green}}$. [4]

$y=|x-2|$ $\quad$ $y=2x-3$

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$y=|x-2|+0$

Vertex

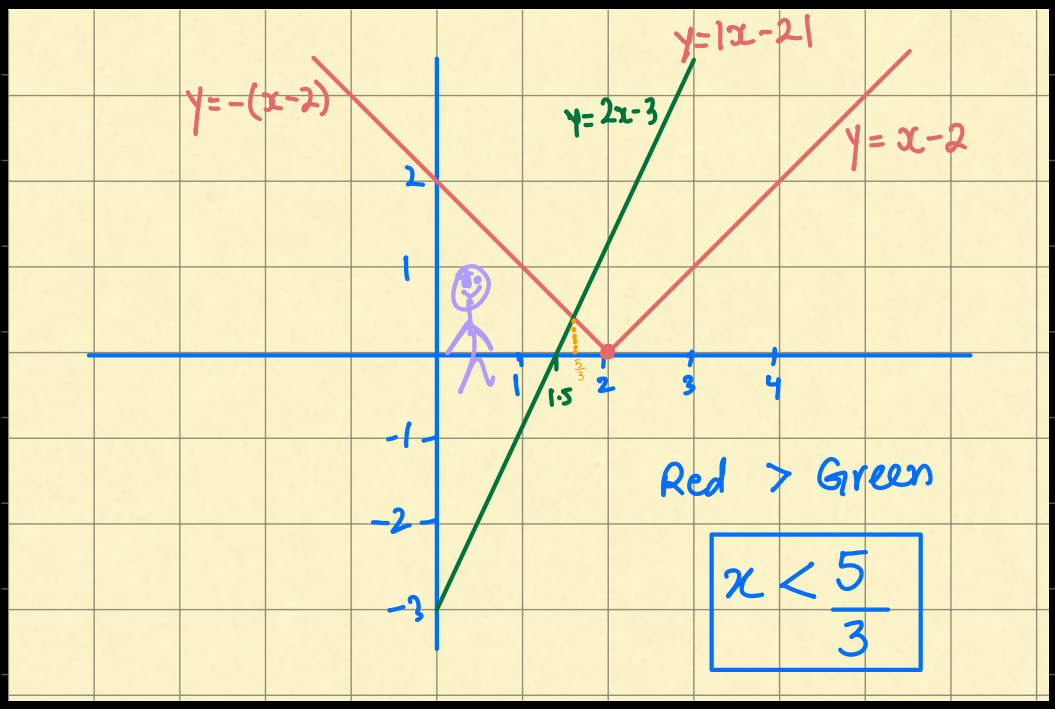
$x=2, y=0$

y-int, $x=0$, $y=|0-2|=|-2|$
$=2$

$y=2x-3$

x-int, $y=0$, $0=2x-3$, $3=2x$, $x=1.5$

y-int, $x=0$, $y=2(0)-3$, $y=-3$



Point of intersection

$y=2x-3$, $y=-(x-2)$

$2x-3=-(x-2)$

$2x-3=-x+2$

$3x=5$

$x=\frac{5}{3}$

Red < Green

18 Solve the inequality $\underbrace{|x-3|}_{y=|x-3|} < \underbrace{3x-4}_{y=3x-4}$. [4]

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$y = |x-3| + 0$

Shape =

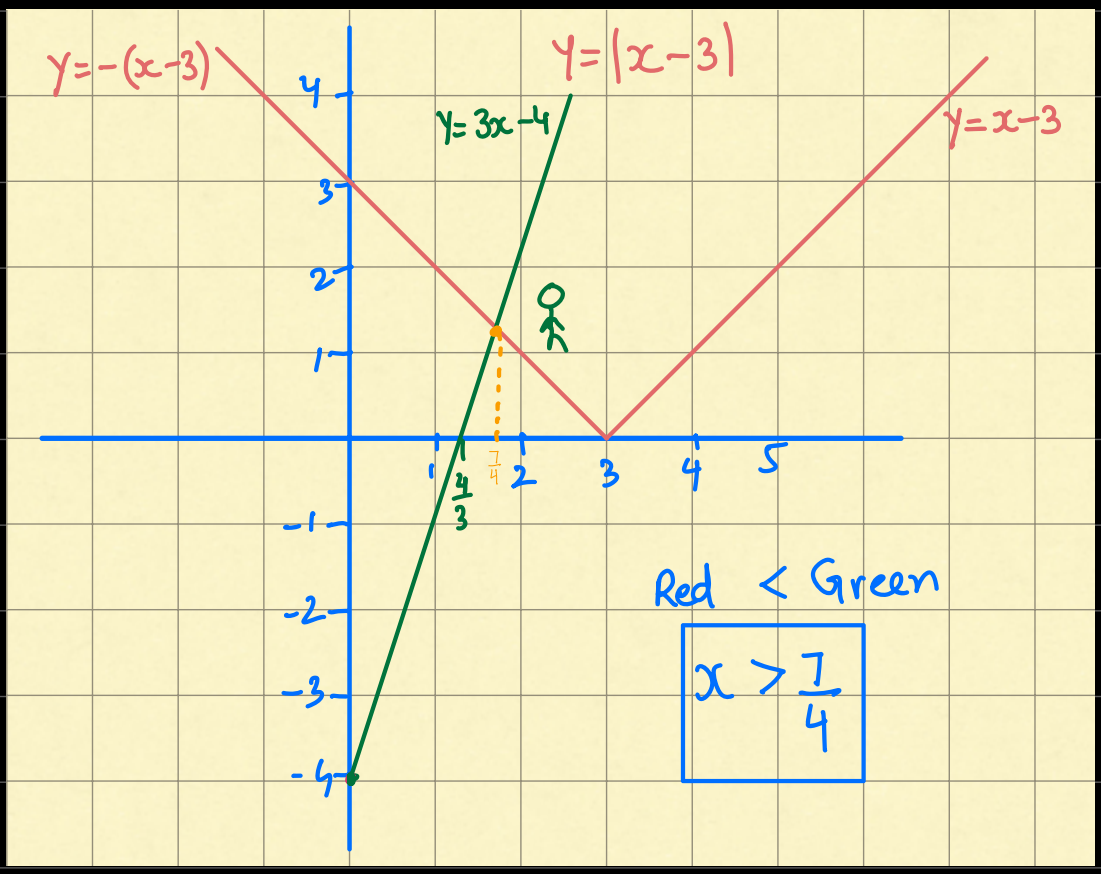
Vertex = $(3,0)$

y-int = 3

$y = 3x - 4$

x-int : $y = 0$: $0 = 3x - 4$, $x = \frac{4}{3}$

y-int : $x = 0$: $y = 3(0) - 4 = -4$



Red < Green

$\boxed{x > \frac{7}{4}}$

Point of intersection

$y = 3x - 4$, $y = -(x-3)$

$3x - 4 = -(x-3)$

$3x - 4 = -x + 3$

$4x = 7$

$x = \frac{7}{4}$

3 Solve the inequality $2x > |x-1|$. [4]

Red > Green

$y = 2x$ $\quad$ $y = |x-1|$

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$y = 2x$

$x = 0, y = 0$

$x = 1, y = 2(1) = 2$

$y = |x-1| + 0$

$|0-1| + 0$

$|0-1|$

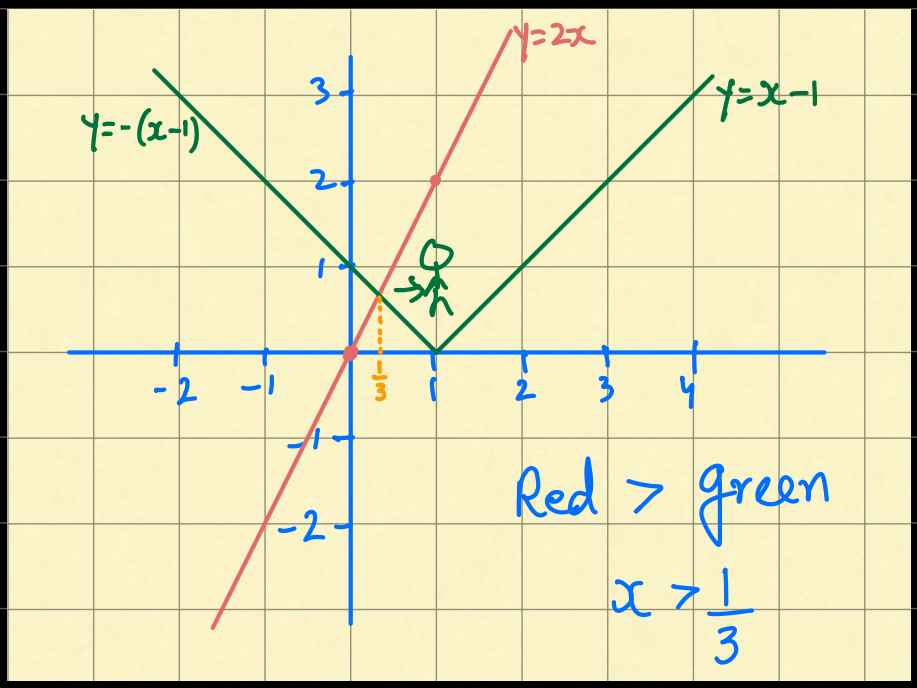
$|-1|$

$+1$

Shape : $\vee$

Vertex = $(1, 0)$

y-int : 1



Point of intersection.

$y = 2x$, $y = -(x-1)$

$2x = -x + 1$

$3x = 1$

$x = \frac{1}{3}$

6 Solve the inequality $|x + 3a| > 2|x - 2a|$, where a is a positive constant. [4]

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$$(|x+3a|)^2 > (2|x-2a|)^2$$

$$(x+3a)^2 > 4(x-2a)^2$$

Solved super!

$\boxed{Q2}$

$$(|x-3a|)^2 > (|x-a|)^2$$

$$\cancel{x^2} - 6ax + 9a^2 > \cancel{x^2} - 2ax + a^2$$

$$9a^2 - a^2 > 6ax - 2ax$$

$$8a^2 > 4ax$$

$$\frac{8a^2}{4a} > x$$

$$2a > x$$

$$x < 2a$$

Q4) Leave it till you do logs.

Q12)

$$(|x+2a|)^2 > (3|x-a|)^2$$

$$(x+2a)^2 > 9(x-a)^2$$

$$x^2+4ax+4a^2 > 9(x^2-2ax+a^2)$$

$$x^2+4ax+4a^2 > 9x^2-18ax+9a^2$$

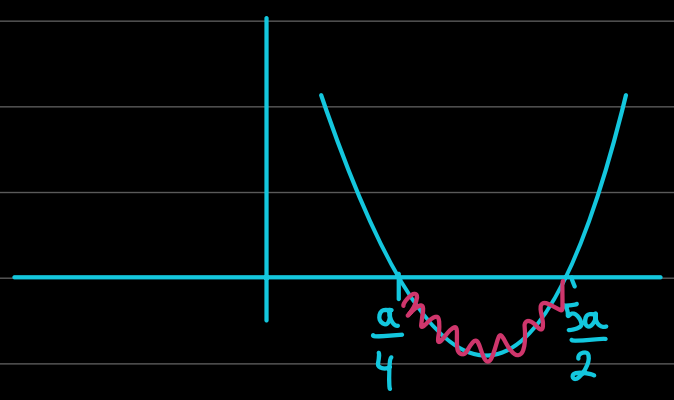
$$0 > 8x^2-22ax+5a^2$$

$$8x^2-22ax+5a^2 < 0$$

$$8x^2-20ax-2ax+5a^2 < 0$$

$$4x(2x-5a)-a(2x-5a) < 0$$

$$(2x-5a)(4x-a) < 0$$



$$\frac{a}{4} < x < \frac{5a}{2}$$

10 Solve the equation $|x-2| = \left|\frac{1}{3}x\right|$. [3]

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Q13) $(|3x-1|)^2 < (|2x+5|)^2$

$(3x-1)^2 < (2x+5)^2$

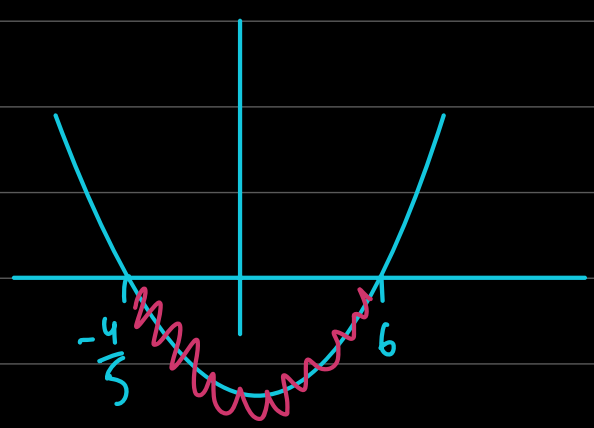
$9x^2 - 6x + 1 < 4x^2 + 20x + 25$

$5x^2 - 26x - 24 < 0$

$5x^2 - 30x + 4x - 24 < 0$

$5x(x-6) + 4(x-6) < 0$

$(x-6)(5x+4) < 0$



$-\frac{4}{5} < x < 6$

Q17) $(|2x+1|)^2 < (3|x-2|)^2$

$$(2x+1)^2 < 9(x-2)^2$$

$$4x^2+4x+1 < 9(x^2-4x+4)$$

$$4x^2+4x+1 < 9x^2-36x+36$$

$$0 < 5x^2-40x+35$$

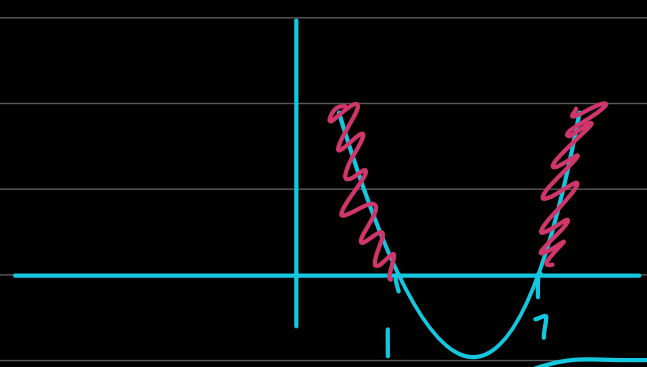
$$\frac{5x^2}{5} - \frac{40x}{5} + \frac{35}{5} > \frac{0}{5}$$

$$x^2-8x+7>0$$

$$x^2-x-7x+7>0$$

$$x(x-1)-7(x-1)>0$$

$$(x-1)(x-7)>0$$



$\boxed{x<1}$ $\boxed{x>7}$

Q19) $(2|2x-a|)^2 < (|x+3a|)^2$

$$4(2x-a)^2 < (x+3a)^2$$

$$4(4x^2-4ax+a^2) < x^2+6ax+9a^2$$

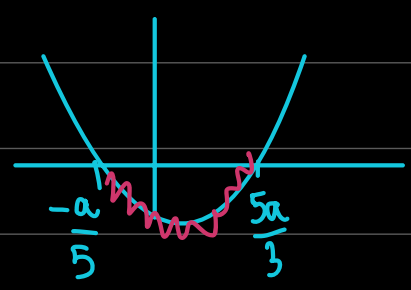
$$16x^2-16ax+4a^2 < x^2+6ax+9a^2$$

$$15x^2-22ax-5a^2<0$$

$$15x^2-25ax+3ax-5a^2<0$$

$$5x(3x-5a)+a(3x-5a)<0$$

$$(3x-5a)(5x+a)<0$$



$$-\frac{a}{5} < x < \frac{5a}{3}$$